Lab Rotation Report: Exploring Cross-Location Generalization on the PanAf-FGBG Dataset

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Abstract

This project investigates methods to improve domain generalization in primate action recognition using the PanAf-FGBG dataset. Wildlife camera trap models frequently suffer from background bias, leading to poor generalization in unseen environments because the models overfit to static background contexts rather than learning the actual action. This research hypothesized that isolating motion by using optical flow as the primary input modality instead of standard RGB could force the model to focus strictly on the action, thereby improving cross-location generalization. The study consisted of extensive experiments comparing optical flow against RGB and late-fusion models on both overlapping and disjoint data splits. Ultimately, the core hypothesis was rejected; pure optical flow underperformed due to the loss of critical fine-grained texture and susceptibility to environmental noise. However, late fusion of RGB and optical flow provided marginal performance improvements over individual modalities.

1 Research Question

Action recognition models in wildlife settings often focus heavily on the background, which yields poor generalization when deployed in new locations. To combat this, we suggest converting RGB videos to optical flow, effectively stripping away the static backgrounds in the videos, leaving only pure motion signals for the models to process. Models that successfully isolate and focus on the actual action demonstrate stronger generalization capabilities across diverse environments and we suggest optical flow modality can help models achieve this. The main research question is: **Can using optical flow as the input modality improve generalization in primate action recognition models?**

2 Dataset

The research utilizes the PanAf-FGBG dataset (Brookes et al., 2025), comprised of remote camera trap footage collected from various conservation sites in 6 African countries. The dataset contains 5,070 sample pairs consisting of isolated foreground and background videos, encompassing 14 distinct primate action classes.

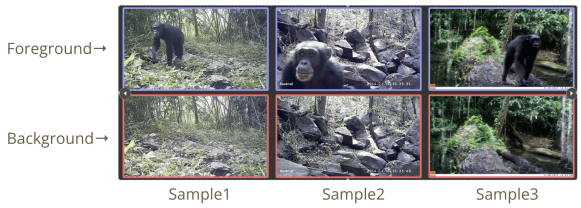


Figure 1: PanAF-FGBG dataset examples.

The class distribution is heavily imbalanced; majority classes include “Travel” and “Resting”, while minority classes include “Playing” and “Piloerection”. [1]

To rigorously evaluate performance, the dataset features two distinct views:

- **Overlapping Split:** The training, validation, and test sets share the same camera locations.
- **Disjoint Split:** The test set contains camera location that are completely unseen in training and validation splits. This serves as the true benchmark for domain generalization. [2]

3 Methodology

The primary task is primate action recognition across variable environmental domains. Model performance is evaluated using two primary metrics: μAP (micro Average Precision), which calculates

[1] See Appendix A.1 for full distribution of classes.
[2] See Appendix A.2 for table with full percentage splits for both data views.

performance across all samples globally (making it easily skewed by frequent classes), and mAP (macro Average Precision), which calculates the metric independently for each class and averages them for a more balanced evaluation.

Brookes et al. (2025) tackled background bias by performing latent space subtraction. They extracted features for both foreground and background videos using vision encoders and subtracted the background features from the foreground features ($F - B$) in the latent space to neutralize the bias. The diagram in Figure 2 shows this approach.

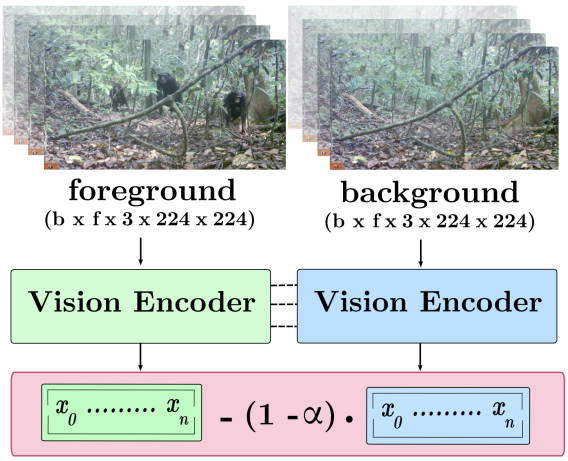


Figure 2: Latent Space Background Compensation (This is Figure 6 in Brookes et al. (2025))

Our approach is to completely ignore the background by isolating motion at the input level. We utilized DPFlow (Morimitsu et al., 2025), a state-of-the-art deep learning-based estimator, to extract pure optical flow from the RGB videos in the dataset.



Figure 3: Example showing a single frame of the same video after the RGB video (Left) is converted to optical flow (Right). [3]

4 Experimental Setup

The architecture relied on pretrained I3D ConvNet backbones (Carreira and Zisserman, 2017) , resulting in two distinct models: I3D Flow and I3D RGB. Models were fine-tuned across both the Overlapping and Disjoint splits. To ensure robustness, extensive data augmentations were applied, including resizing, noise injection, horizontal flips, random cropping, and temporal cropping.

[3]Optical flow was visualized using the technique introduced by (Baker et al., 2011).

5 Results & Key Findings

The quantitative results of our experiments are summarized in Table 1.

	Overlapping		Disjoint	
Model	μAP	mAP	μAP	mAP
Baseline (FG-Only)	72.8	49.3	**57.7**	**30.9**
Our I3D (Flow)	60.1	34.8	46.5	24.4
Our I3D (RGB)	74.3	52.0	51.8	26.7
Late Fusion	**75.2**	**52.7**	53.7	28.5

Table 1: Comparison of model performance across data splits.

The Baseline (FG Only) is (Brookes et al., 2025)'s scores for models trained on only the foregorund videos. Therefore these scores contain background bias and we want to see whether our approach achieved better scores indicating a reduction in background bias.

Our I3D (Flow) model significantly underperformed compared to the baseline. Through superior hyperparameter tuning, our I3D (RGB) model outperformed the baseline on the Overlapping split (μAP 74.3), though it still struggled with domain generalization on the Disjoint split. Fusing the RGB and Flow modalities improved all of our model scores, yielding an Overlapping μAP of 75.2 and a Disjoint μAP of 53.7.

Key Findings: The core hypothesis that pure optical flow improves domain generalization in primate action recognition is rejected. Results indicate that standard RGB remains a superior primary modality for this domain.

6 Discussion & Future Work

The failure of pure optical flow as a standalone modality can be attributed to three main factors:

1. Loss of Fine-Grained Texture: Complex actions such as "feeding" or "grooming" rely heavily on delicate hand-object interactions that require fine-grained spatial textures entirely absent in raw optical flow.

2. Noisy Motion Fields: In wild environments, external factors like wind or camera shake

generate spurious motion vectors that severely contaminate the optical flow field.

3. Appearance as Context: While background surroundings introduce bias, static appearance cues located directly on the primate's body such as posture, relative size, and lighting interactions carry crucial semantic clues required for accurate classification.

Future work should explore advanced foundation models to better disentangle subject from background. Specific future directions include integrating the DinoV2 modality into the I3D architecture and transitioning to Vision Transformer (ViT) backbones to better leverage robust, context-aware feature representations.

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Disclosure: LLMs were used to troubleshoot BibTeX and LaTeX.

References

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A Appendix

A.1 Class Distribution in PanAf-FGBG dataset

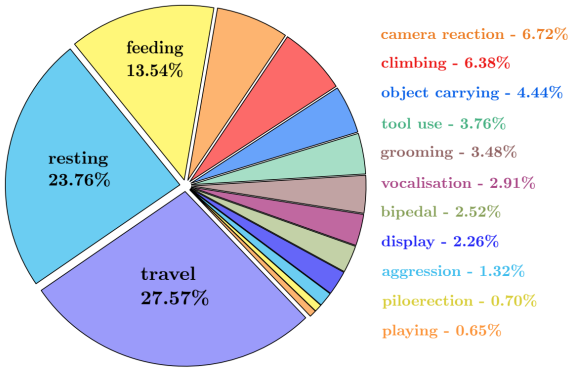


Figure 4: PanAF-FGBG dataset class distribution. (This is Figure 2 in (Brookes et al., 2025))

A.2 Data Split Percentages in panAF-FGBG

Table 2: Dataset Split Statistics

	$D_{\text{Overlapping}}$		D_{Disjoint}	
Split	Videos	%	Videos	%
Train	2,847	56%	2,782	55%
Validation	1,221	24%	1,219	24%
Test	1,002	20%	1,069	21%
Total FG-only	5,070	100%	5,070	100%

Table 3: Dataset Split Statistics